



The Demoscene

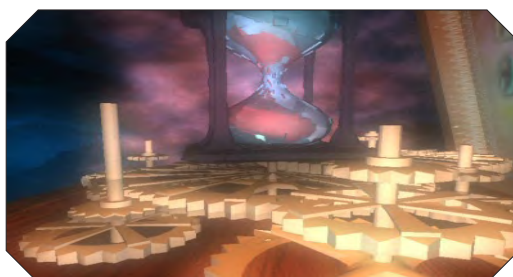
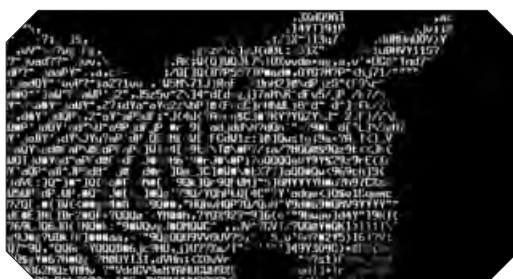
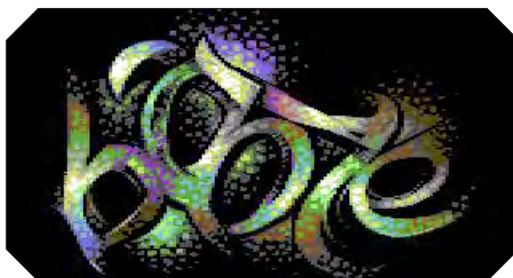
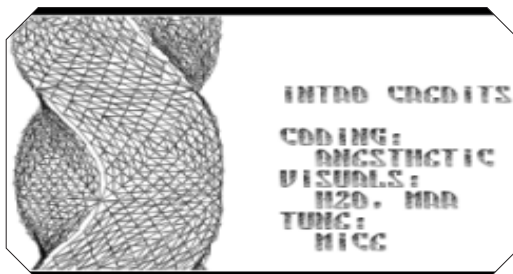
From intros for cracked software, to an art movement, we trace the evolution of the mostly underground demoscene.

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For most people, a “demo” means a limited version of a program that you can try out before making a purchase. However, if you belong to a mostly scattered community of elite programmers and artists, a “demo” is a very versatile real-time audio-visual presentation. The content of the presentation varies drastically, typically involving spectacular effects, synthesised music and an overall design. Demos are usually programmed by a group of people, and distributed over forums and web sites. The demoscene has a history dating back to the 1980s, when personal computers were first made available to the masses. Since then, the movement has been lurking in the shadows, neither growing nor fading, but changing in style and method.

The C64, Amiga and Atari consoles were among the first machines used to program demos. Original software was costly back then, and users traded disks with cracked versions of programs. The people responsible for cracking the software, tagged the cracked versions with pseudonyms for identification that helped in making sure of authenticity, an early version of the tags used with torrent files today. What began as a simple image or a macro to identify the cracked versions of the software grew to become elaborate sequences of sounds and images. These sequences were intros to cracked versions, called “cracktros”. Since the software took up most of the space on a disk, there was very little leeway for the crackers to work with. The fallout was that demo artists had to create the most spectacular effects, designs and sound possible, and fit them in a tiny space. Even today, the smaller the demo, the greater the respect it commands. A 64-kB sequence, complete with generated effects and music, can take up as much as 20 MB in a modern FLV format encoded for easy distribution on the internet.

One of the key aspects of programming demos was that everything was calculated by the computer in real time and graphics was rendered on the fly. Demos pushed the machine to new limits, and took weeks, if not months, of programming. For this reason, groups or teams emerged, called a “demogroup”. A typical demogroup consisted of anywhere between four to ten individuals, who each performed a specific task. The cracker



broke down the copy protection and other legal restrictions on a program. An artist, typically from some other underground faction such as a graffiti artist designed the logos and the graphics of the demo. Perhaps another artist would convert these designs to a digital format. A music artist programmed the music, which would typically be some form of electronica or industrial sounding track. A swapper was responsible for spreading the demo, maintaining activity on forums, and most importantly, making all the contacts that would spread the software over the internet. All of this would come together on a single disk as a demo and a cracked program. Most of these people were anonymous, known only through their monickers.

Many of these groups were organised across wide geographic locations. Some of them had cycling members, and included over 100 people working at any point of time. This involved a large number of crackers, and a large number of traders or swappers. This allowed for the latest software to be quickly distributed to a large number of people. The demoscene was and is active mostly in Europe, particularly in the Scandinavian nations. 1985 to 1995 is considered by many in the scene to be the golden age of the demo, when the demos flourished the most. There are many reasons for this perception. Before the Windows-based PC came along, Amiga, Commodore and Atari were the computers used by most people. The hardware capabilities of these machines were more or less similar, and upgrades were almost unheard of. The components were fixed, and there was no need to cycle them as newer versions came into the market. All these consoles outperformed the early Windows-based machines in terms of native graphics and audio capabilities. Graphic cards and audio cards were very rare, and VGA was yet to make its presence felt. This allowed for the first generation of demo programmers a level playing field, where innovation was very hard to achieve.

More than benchmarks, software and perhaps even games, demos were responsible for pushing machines to their absolute limits. Demo programmers exploited every bug and glitch in the machine, both known and unknown, to program that many more components into the demo. The way a machine operated was studied and used in every conceivable way—the swap space, the